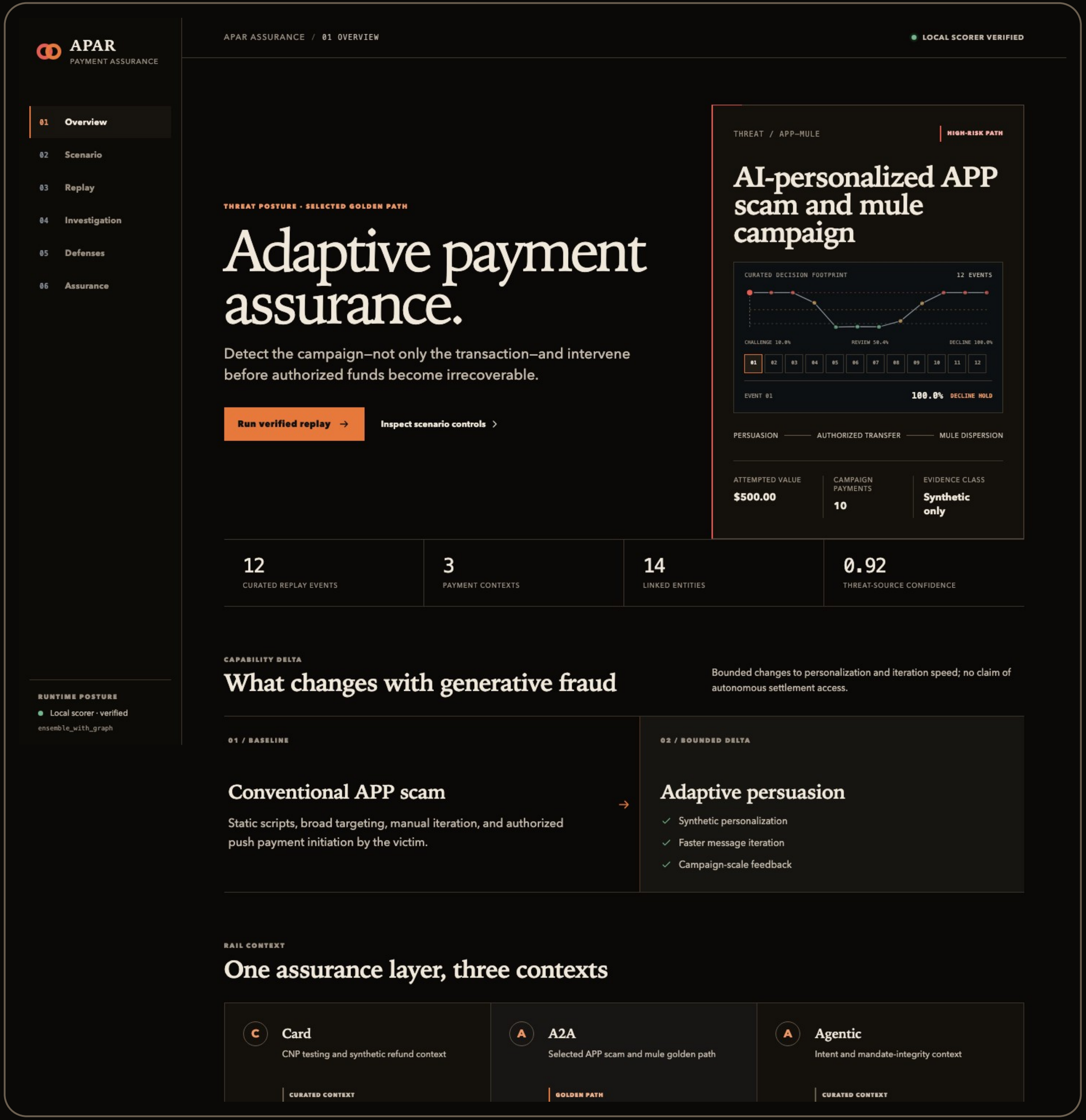
MASTERCARD INNOVATION CHALLENGE 2026

Assure the campaign, not just the transaction.

APAR turns emerging GenAI payment threats into rail-correct synthetic campaigns, tests layered defenses, and preserves a human promotion boundary.

SYNTHETIC • OFFLINE • REPLAYABLE

Dylan Moraes • Competition submission



GenAI changes the operating model of fraud

01

PERSONALIZE

Victim-specific content at scale

→

02

ITERATE

Faster adaptation to feedback

→

03

COORDINATE

Many entities, rails, and time windows

→

04

DELEGATE

Agents can act on user intent

Transaction classifier → Campaign-aware assurance system

Behavior + lifecycle + economics + authority + governance

One assurance loop connects five control planes

IDENTIFY		GENERATE		DEFEND		VERIFY		GOVERN
Evidence + sources threat registry	→	Rail + ledger-correct campaigns	→	Rules + calibrated graph ensemble	→	Identity + intent integrity	→	Controls + human promotion

Bounded feedback returns failed controls to the research backlog—never directly to deployment.

Failed experiments shaped the winning architecture

V1

Workload cap failed

1.7857% review floor
vs 1% cap

V4

Missing metrics failed closed

Calibration + alert time
were not invented

V5a

Perfect score rejected

Future graph leakage
removed and recorded

V5b

Real rail evidence

Generator → simulator →
rail → ledger → row

V5c

Graph arm selected

Best frontier;
full hybrid rejected

Novelty: not a single algorithm—an evidence-governed loop that learns from its own failure modes.

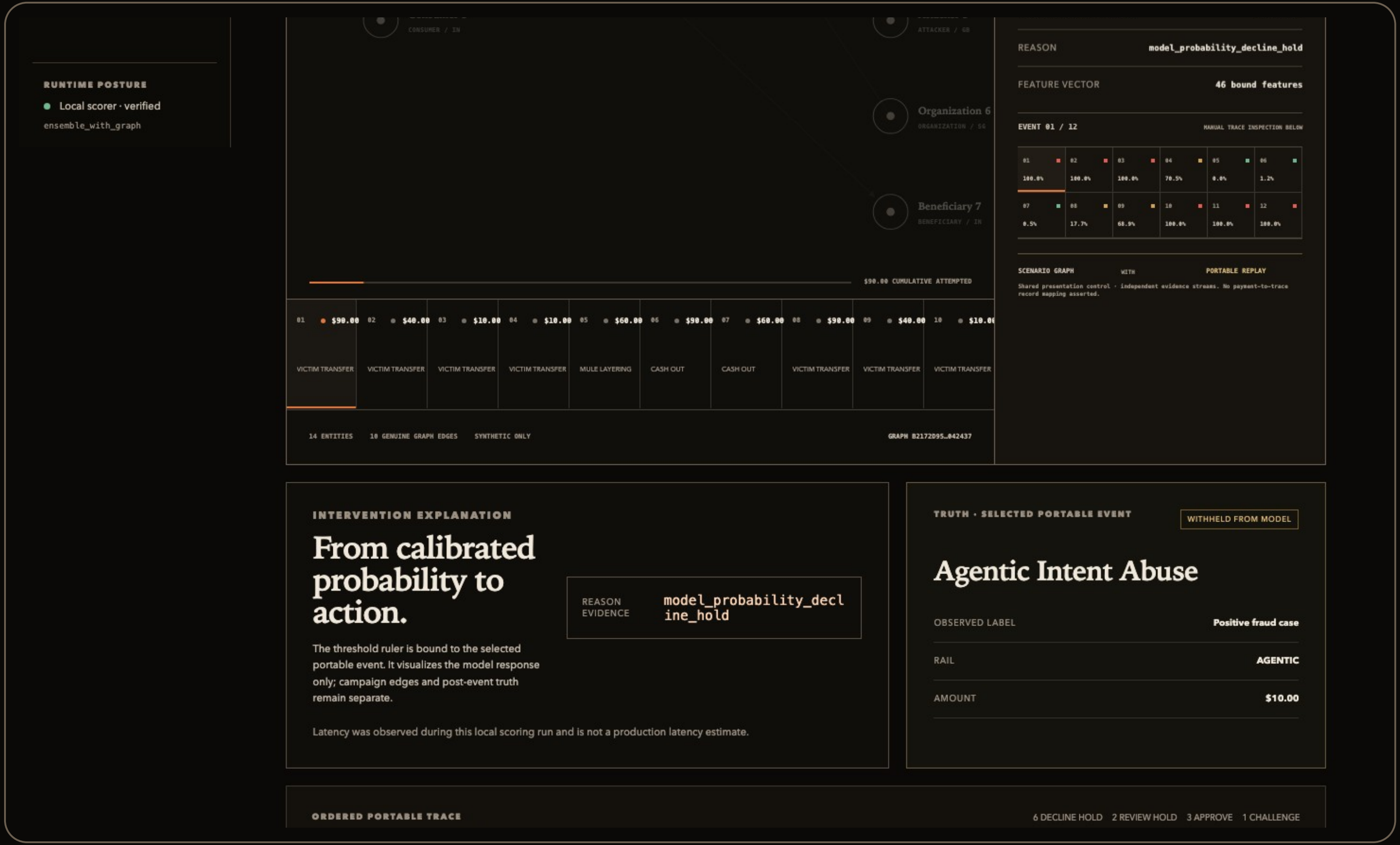
Graph context wins the operational frontier

Verified recovered diagnostics — synthetic and non-authoritative

ARM	RECALL %	PRECISION %	F1 %	FALSE-DECLINE %	P95 MS
Rules only	85.95	14.88	25.37	87.44	4.06
Ensemble • no graph	99.74	94.76	97.19	0.000	3.38
Ensemble • graph	**99.87**	**95.88**	**97.83**	**0.0037**	**3.54**
Full hybrid	99.93	16.85	28.83	87.44	19.74

Precision +1.12 pts **Challenge -0.23 pts** **Latency +0.16 ms p95**

A small, portable model powers the live demo



3 ×	CatBoost members
46	frozen causal features
12	hash-bound scenarios
0.0	max replay error

Approve • Challenge • Review hold • Decline hold

Investigators see one campaign—not twelve alerts

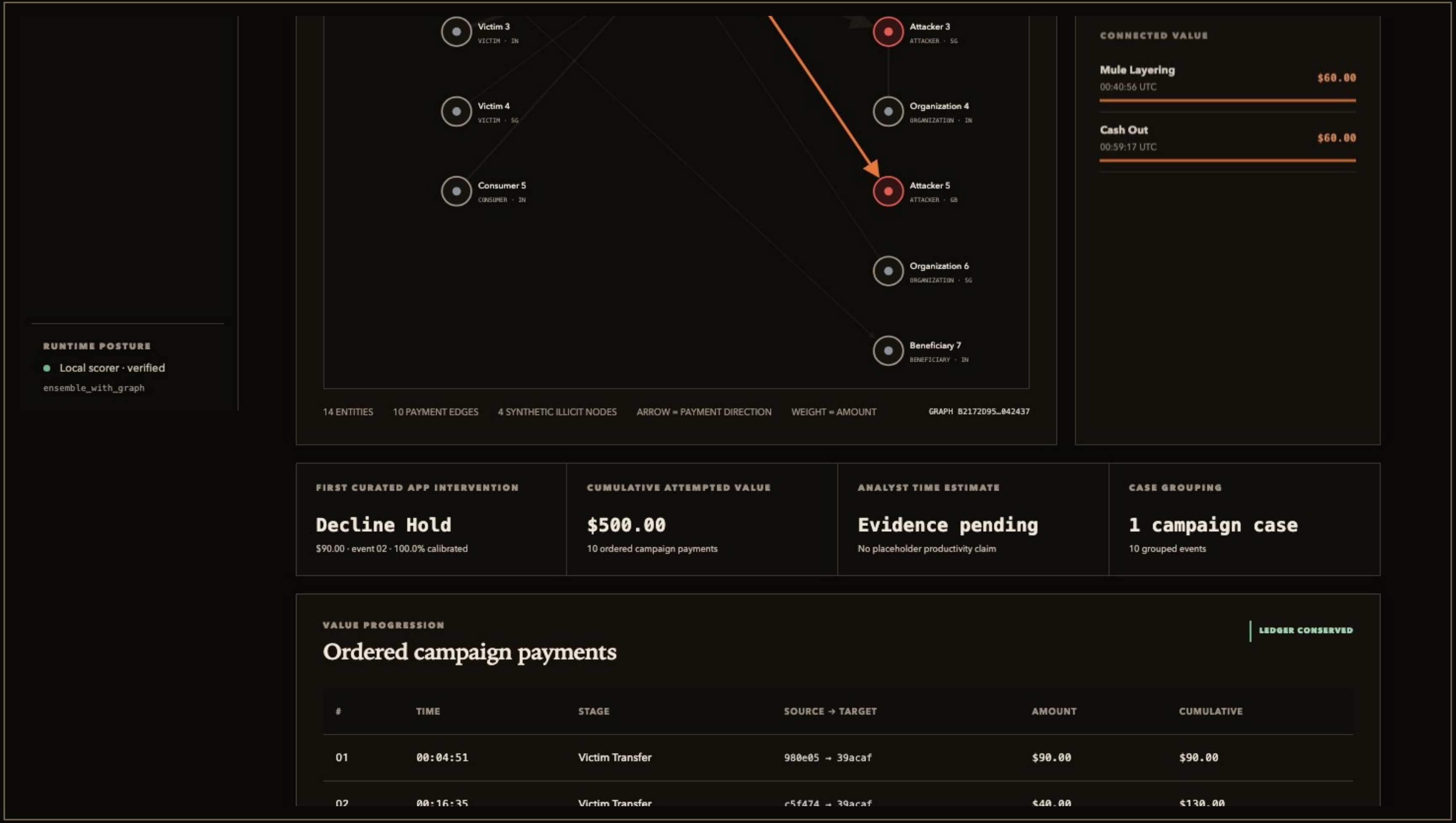
CASE-CENTRIC

14
entities

10
payment edges

$500
conserved synthetic ledger

Analyst-time benefit: evidence pending



Agentic commerce needs proof before probability

- Agent identity
- Merchant + cart
- User mandate
- Expiry + nonce
- Scope + amount
- Replay rejection

DEFINITIVE FAILURE

DECLINE_HOLD before statistical risk

SEPARATE PROOF POINT · AGENTIC INTEGRITY

TrustVerifier rejects invalid authority.

Identity, mandate, scope, cart binding, and replay checks are genuine tested controls. This is a deterministic authorization proof—not evidence of graph-model performance.

TEST SOURCE

`tests/trust/test_verifier.py`

01	Identity	AGENT_IDENTITY_MISMATCH	✓ TESTED
02	Mandate	MANDATE_SCOPE_VIOLATION	✓ TESTED
03	Scope	AMOUNT_LIMIT_EXCEEDED	✓ TESTED
04	Binding	CART_HASH_MISMATCH	✓ TESTED
05	Replay	NONCE_REPLAY	✓ TESTED

Evidence governance is part of the product

01

Causal features

Future append + equal-time isolation

02

Executed controls

Shuffle, rename, leakage, single-class

03

Immutable evidence

Models, traces, metrics, manifests hashed

04

Independent replay

Fresh environment re-scores all 12 cases

05

Human promotion

The model cannot approve itself

Portable bundle

52ed4c31...dbc900

Promotion boundary

Human approval required

NO SELF-PROMOTION

A sharp claim boundary makes the demo credible

WHAT WE PROVE

- ✓ Portable model loads and replays exactly
- ✓ Four rail-backed synthetic families
- ✓ Graph/no-graph architecture comparison
- ✓ Deterministic TrustVerifier integrity
- ✓ Offline console + hash-bound fallback

WHAT WE DO NOT CLAIM

- × Production or real-cardholder performance
- × Production-readiness or scale evidence
- × External or multi-institution validation
- × Autonomous model promotion
- × Claims beyond the demonstrated model

Synthetic evidence before assertion.

From competition prototype to governed payment assurance

1

OFFLINE REPLAY

Authorized historical samples

→

2

SHADOW MODE

Live features; no authority

→

3

ASSISTED OPS

Human-controlled friction

→

4

CHALLENGER

Limited traffic + rollback

→

5

GOVERNED SCALE

Continuous adversarial regression

APAR

Assure the campaign. Verify the intent.
Promote only the evidence.

DEMO READY